People and Earth's Ecosystem
MS. CAMILLE ACE NAVACILLA

THE DOUBLE-EDGED SWORD:
THE HUMAN STORY OF MINING

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INTRODUCTION

Mining is a crucial economic activity that supplies the raw materials required for contemporary infrastructure, technology, and energy, including metals, coal, and minerals. Nevertheless, the procedure is essentially harmful, requiring the physical relocation of soil and the application of expensive chemical procedures. Significant environmental degradation on a local, regional, and worldwide scale may arise from these practices.

Positive Impacts

The mining sector, which is sometimes disregarded and underestimated, is crucial to the modern world. Without mined metals and minerals, technological innovations like computers, smartphones, and infrastructures would not be possible. Similarly, metals like nickel and copper are essential for the deployment of electric vehicles and the integration of renewable energy into the national grid as the nation moves toward cleaner and more efficient energy.

In communities

One of the world's most mineral-rich archipelagos is home to the Philippines. For generations, both domestic and foreign interests have been fascinated by the abundance of gold, copper, nickel, chromite, and coal that runs beneath its mountains and islands. In the past, mining has been one of the few methods out of the harsh rural poverty experienced by communities in the uplands of Benguet, Bukidnon, and Surigao. The mine is frequently the largest employer within a hundred kilometers in towns close to large-scale activities. In areas where the only options are seasonal farming or migrant employment overseas, miners' poor wages by Manila standards signify a change. Through the national government's financial allotment system, local governments in mining municipalities receive a portion of the proceeds, which they use to build classrooms, barangay health centers, and farm-to-market routes that link remote towns to the larger economy. Revenues from the nickel mine have helped Surigao del Norte create infrastructure that would have otherwise taken decades. Families in those communities do not view the mine as an abstract concept. It is the reason the school has a roof.

In society

Beyond the localities where extraction takes place, Philippine mining has an impact. Every Filipino with a smartphone has copper and cobalt in their pockets; these minerals are partially generated by local ore. Mined resources are used in the nation's manufacturing, electronics, building, and power grid industries. Stainless steel, which is utilized in hospitals and food processing facilities nationwide, contains nickel. What comes out of the earth is directly related to the larger narrative of industrialization—the slow but uneven process by which a developing country constructs the physical infrastructure of contemporary life. Advocates of mining argue, and with good reason, that the sector finances the present while enabling the future. In a nation where millions

of people still lack access to clean running water and dependable energy, and where the minerals required to address those issues are found beneath Filipino soil, the argument is powerful.

Negative Impacts

Climate Rights International stated in a new report and video released today that the Philippine government, nickel mining companies, and downstream users of nickel, such as electric vehicle (EV) and battery companies, must act immediately to prevent and correct environmental degradation and human rights abuses resulting from nickel mining. The Philippines is the world's top exporter of raw nickel ore, a material used in energy storage and electric vehicle batteries, and the world's second-largest producer of nickel.

In an ecosystem

Millions of tons of mine waste were dumped into the Boac River, one of Marinduque Island's main water sources, in 2000 due to a tailings disaster by the Marcopper Mining Corporation. The river was orange in color. Fish passed away. Harvests failed for farmers who relied on the river for irrigation. Long after cleanup efforts were deemed finished, residents continued to report skin rashes, respiratory issues, and a persistent fear of their own water supply. Twenty years later, environmentalists in Marinduque claim that the river has never fully recovered and that the firm that caused it has long since withdrawn, leaving the devastation as a permanent aspect of the island's topography. This tale is reminiscent of what occurred on Rapu-Rapu Island in Albay in 2005, when two leaks from a gold and copper mine destroyed the lives of coastal communities who were not involved in the mine's earnings and killed fish in neighboring waterways.

In communities

Communities having the least ability to oppose mining are frequently the ones most negatively impacted by it. Indigenous peoples, such as the Mangyan in Mindoro, the Igorot groups in the Cordilleras, and the Lumad communities in Mindanao, have seen their ancestral lands become the most sought-after locations for exploration permits. Before any mining operation starts on their land, indigenous tribes must give their free, prior, and informed consent, according to the Indigenous Peoples' Rights Act of 1997. In reality, incomplete consultations, bribes to individuals of the community who are not representative, or outright intimidation have occasionally been used to fabricate such approval. Some Lumad villages in Mindanao have reported intimidation, forced

relocation, and in the worst situations, bloodshed, as a result of their opposition to mining operations. There is a legal framework. In many situations, there is a significant gap between the framework and lived reality. The phrase "consent" loses most of its significance when a community's objection to a mine becomes dangerous to voice.

Reflection

Although it is difficult to resolve, some people honestly manage the conflict at the core of Philippine mining, which is between the wealth it generates and the harm it causes. When enforced, the environmental requirements established by the Mining Act of 1995 and later administrative decrees provide a true floor of protection. In addition to sponsoring community development initiatives that go beyond the mine's lifespan, certain businesses in the nation have made significant investments in rehabilitation programs, planting trees and repairing soil on mined-out ground. The problem is consistency. Historically, enforcement has fluctuated based on business size, political connections, and the volume of noise a given neighborhood can produce. In the Philippine context, responsible mining is not impossible. It is a standard that is a lot neglected and only occasionally met.

Conclusion

What Mankayan, Marinduque, and the Lumad communities of Mindanao have in common is that they are areas where the prosperity of the rest of the nation has been partially borrowed against, and the debt has not always been fully paid. The fisherman in Marinduque and the miner in Benguet are subtly linked to the Filipino family in Manila who are charging a phone. This relationship is a reason to be honest about what is taken, what is left behind, and who is entitled more than they have received. It is not a reason to feel guilty. What the mountain gives is not retained. Whether those who live in its shadow are permitted to is the question.

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